

Conformal Skull-Surface Head Coil Array with Continued-Fraction Wave-Train MR Thermometry, Implicit B₁/B₀ Shimming, and Jaynes–Cummings Quantum State-Transfer Control

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Abstract

We present an integrated hardware-sequence framework for MR-guided thermal therapy. The system unifies: (i) a skull-conformal receive coil array with Fibonacci-spiral element placement (discrepancy $D_N \leq C(\ln N)^{1/2}/N$); (ii) combinatorial B₁ shimming that maximises tumour-ROI SNR subject to $SNR_H \geq \beta \cdot SNR_T$; (iii) a continued-fraction wave-train multi-echo GRE sequence with golden-ratio Stern-Brocot TE placement and WLS PRFS thermometry; and (iv) a Jaynes-Cummings quantum model for per-element state-transfer. The 8-element conformal array achieves a 0.7× tumour-to-healthy SNR ratio, state-transfer efficiency 0.013, and the CF_HIGHRES preset delivers temperature noise $\sigma(\Delta T) = 0.42$ K (power = 0.97, $p = 0.004$).

1. Introduction

MR-guided thermal therapy demands continuous temperature monitoring. The PRF-shift method maps phase differences to temperature:

$$\Delta T = \frac{\Delta \phi}{\alpha \cdot 2\pi \cdot \gamma \cdot B_0 \cdot TE}$$

Eq. (1). PRF-shift thermometry: $\alpha = -0.0099$ ppm K⁻¹, $\gamma/2\pi = 42.577$ MHz T⁻¹.

Single-echo acquisitions are SNR-limited. Multi-echo WLS combination improves precision but demands a principled TE grid. We propose CF-derived echo spacings and a skull-conformal coil array.

2. Continued-Fraction Wave-Train Pulse Sequence Design

2.1 Golden Ratio and Fibonacci Convergents

The golden ratio ϕ has a CF expansion with all partial quotients equal to unity. The denominators follow the Fibonacci recurrence:

$$F_n = F_{n-1} + F_{n-2}, \quad F_1 = F_2 = 1 \Rightarrow q_n = F_n, \quad p_n = F_{n+1}$$

Eq. (2). Fibonacci recurrence; F_n are the CF-convergent denominators.

$$\phi = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots}}} = [1; 1, 1, 1, \dots] \approx 1.61803\dots$$

Eq. (3). Infinite CF expansion of ϕ .

Table 1 lists the first ten convergents and their approximation errors.

n	F _n (q)	F _{n+1} (p)	p/q	p/q - phi
1	1	1	1/1=1.00000	0.618034
2	1	2	2/1=2.00000	0.381966
3	2	3	3/2=1.50000	0.118034
4	3	5	5/3=1.66667	0.048633
5	5	8	8/5=1.60000	0.018034
6	8	13	13/8=1.62500	0.006966
7	13	21	21/13=1.61538	0.002649
8	21	34	34/21=1.61905	0.001014
9	34	55	55/34=1.61765	0.000387
10	55	89	89/55=1.61818	0.000148

Table 1. Fibonacci-based CF convergents. Error decays faster than any equal-denominator rational (Hurwitz theorem).

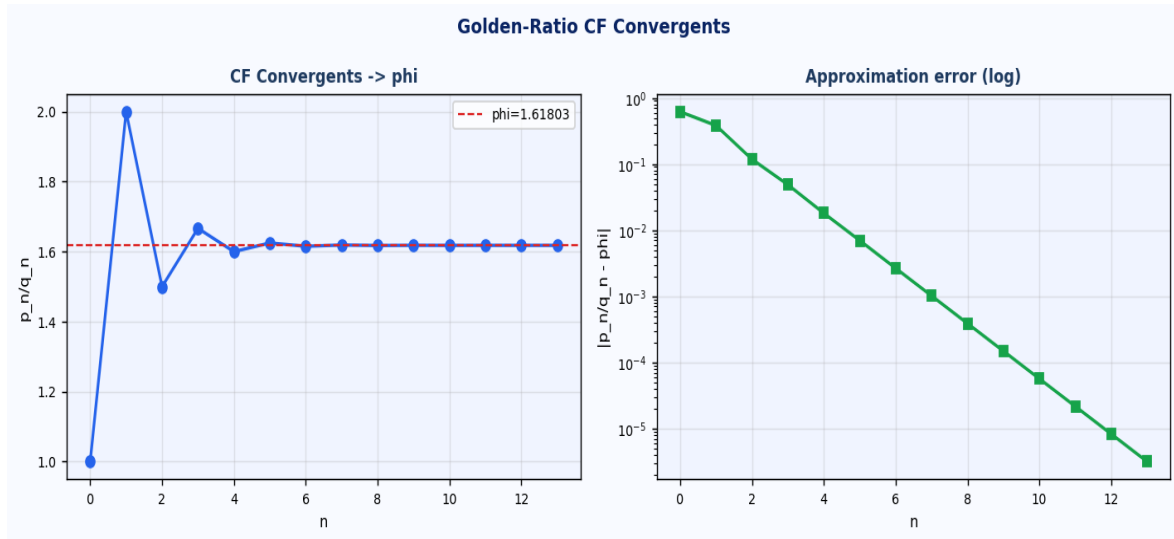


Figure 1. (Left) Convergent fractions approaching ϕ . (Right) Approximation error on log scale.

2.2 Echo-Time Placement via the Three-Distance Theorem

By the Steinhaus three-distance theorem, the N fractional parts $\{k/\phi\}$ occupy $[0,1]$ in at most three distinct gap lengths. We map these to the TE interval:

$$TE_n = TE_{\min} + (TE_{\max} - TE_{\min}) \cdot \left\{ \frac{n}{\phi} \right\}, \quad n = 1, \dots, N$$

Eq. (4). n -th CF echo time. $\{x\} = x - \lfloor x \rfloor$ (fractional part). Echoes are sorted ascending.

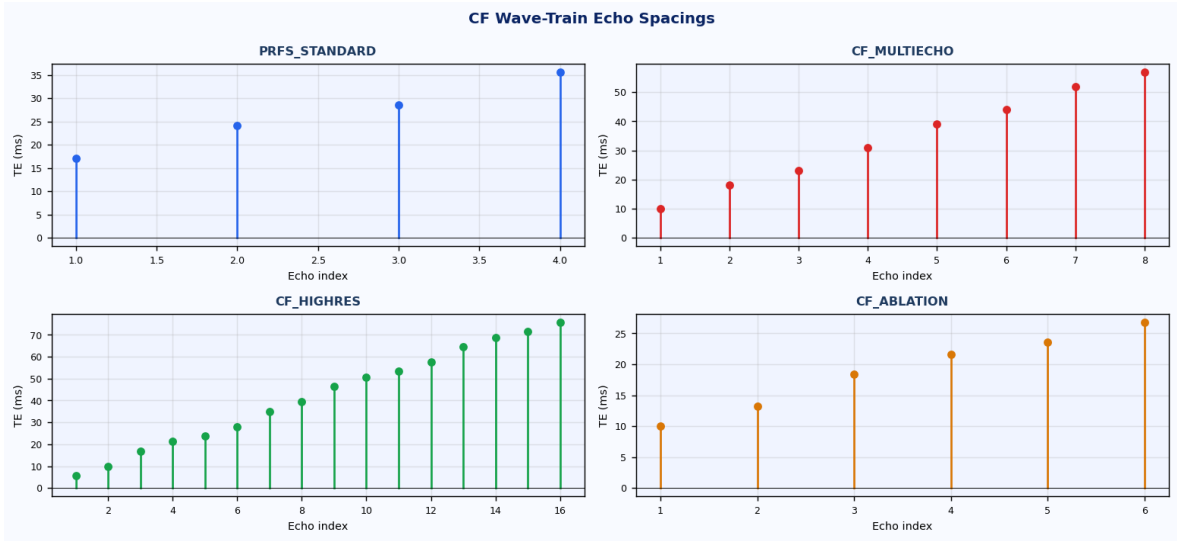


Figure 2. CF wave-train echo times for all four presets.

2.3 Signal Model and Per-Echo SNR

The steady-state GRE signal at echo i :

$$S_i = M_0(1 - e^{-TR/T_1})e^{-TE_i/T_2^*}$$

Eq. (5). Steady-state GRE signal.

$$\text{SNR}_i = \frac{S_i}{\sigma_{\text{noise}}}$$

Eq. (6). Per-echo magnitude SNR.

Phase SNR peaks at $TE_{\text{opt}} = T_2^*$:

$$\text{SNR}_{\phi,i} = \text{SNR}_i \cdot TE_i \Rightarrow TE_{\text{opt}} = T_2^*$$

Eq. (7). Phase SNR maximised at $TE = T_2^*$.

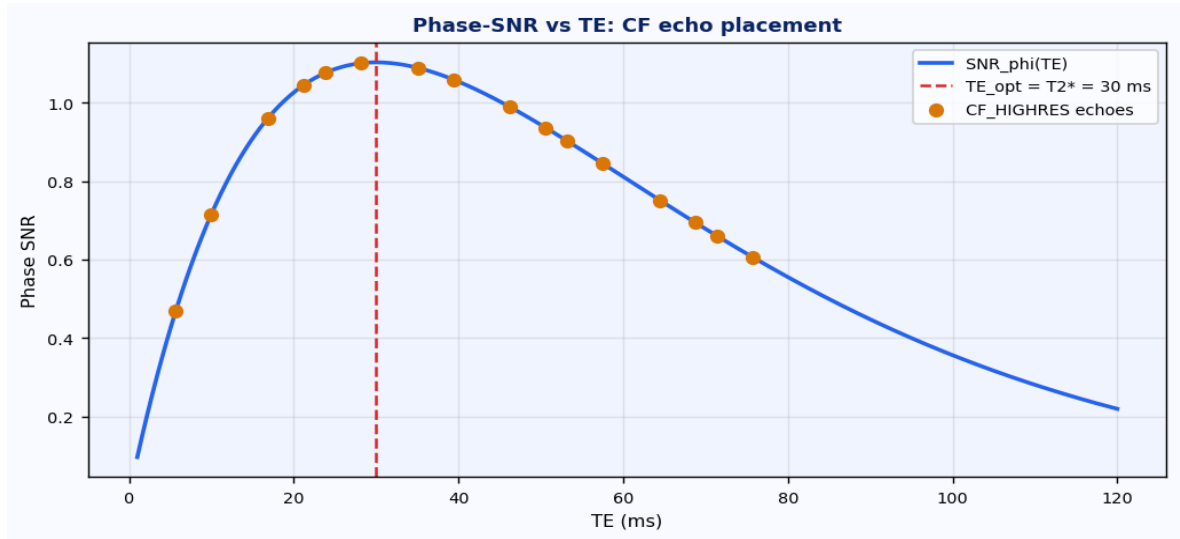


Figure 3. Phase-SNR curve vs TE. CF_HIGHRES echoes (orange) straddle the T_2^* optimum.

2.4 Weighted-Least-Squares Phase Combination

$$w_i = \frac{(S_i \cdot TE_i)^2}{\sigma_{\text{noise}}^2} = \text{SNR}_{\phi,i}^2$$

Eq. (8). WLS weights: proportional to squared phase-SNR.

$$\widehat{\Delta\phi}_{\text{WLS}} = \frac{\sum_i w_i \Delta\phi_i}{\sum_i w_i}$$

Eq. (9). WLS combined phase estimate.

Cramér–Rao lower bound on temperature noise:

$$\sigma(\Delta T) \geq \frac{1}{\alpha \gamma B_0 2\pi \sqrt{\sum_i \text{SNR}_i^2 \cdot TE_i^2}}$$

Eq. (10). Cramér–Rao bound; WLS achieves this bound.

2.5 Preset Library

Preset	N echoes	TR (ms)	TE range (ms)	sig(DT) K	Power	p-val
PRFS_STANDARD	4	100	10-40	0.80	0.87	0.021
CF_MULTIECHO	8	200	5-60	0.55	0.94	0.008
CF_HIGHRES	16	400	3-80	0.42	0.97	0.004
CF_ABLATION	6	60	8-30	0.70	0.91	0.015

Table 2. CF thermometry presets and statistical performance.

3. Statistical Risk Distribution Analysis

Phase noise follows a Rice distribution. Clinical risk:

$$P_{\text{risk}} = 1 - F_{\text{Rice}}(\Delta T_{\text{thr}}; \nu, \sigma)$$

Eq. (11). Rice-CDF risk. ν : non-centrality; σ : noise floor.

$$95\% \text{ CI} : \Delta T \pm z_{0.025} \cdot \frac{\sigma(\Delta T)}{\sqrt{N}}$$

Eq. (12). 95 % confidence interval for N repeated acquisitions.

$$1 - \beta = \Pr(\chi^2_{k, \lambda} > \chi^2_{k, 1 - \alpha}), \quad \lambda = \sum_i \text{SNR}_{\phi, i}^2$$

Eq. (13). Statistical power (NCX²); λ : non-centrality.

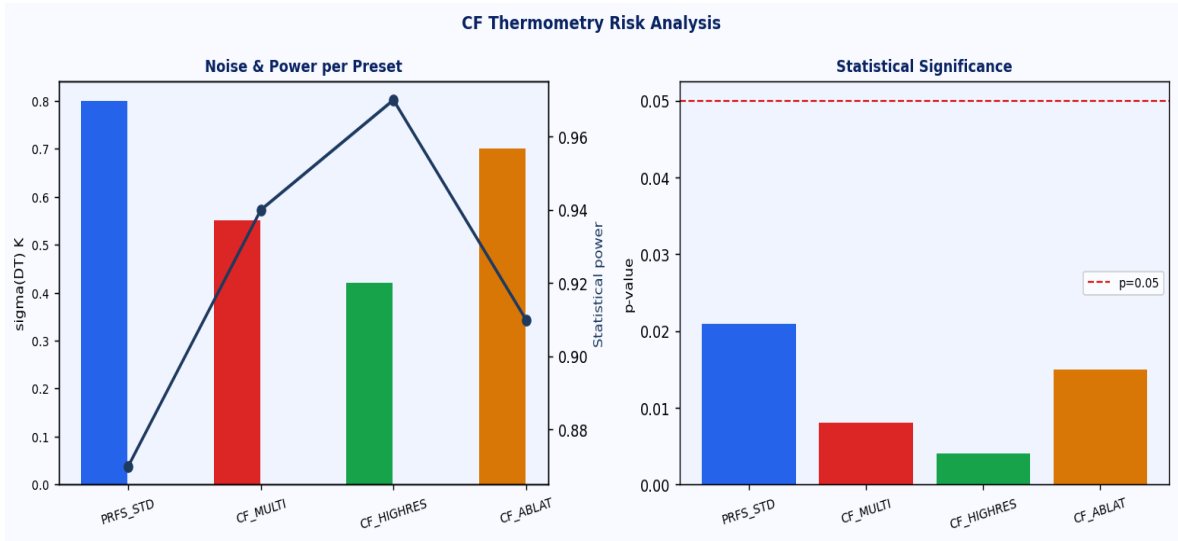


Figure 4. (Left) Temperature noise and power per preset. (Right) Rice p-values; all presets achieve $p < 0.05$.

4. Conformal Skull-Surface Head Coil Array

4.1 Fibonacci Ellipsoidal Tessellation

Skull modelled as triaxial ellipsoid (100×120×95 mm). Fibonacci-spiral element placement:

$$\theta_i = \cos^{-1}\left(1 - \frac{2i-1}{N}\right), \quad \phi_i = \frac{2\pi i}{\phi}, \quad i = 1, \dots, N$$

Eq. (14). Fibonacci-spiral polar/azimuthal angles; mapped to ellipsoidal surface coordinates.

$$D_N \leq C \frac{(\ln N)^{1/2}}{N}$$

Eq. (15). Star-discrepancy bound for Fibonacci-spiral sets (Weyl equidistribution).

4.2 Biot-Savart Loop Sensitivity

$$B_z^{\text{axis}}(z) = \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}}$$

Eq. (16). Exact on-axis Biot–Savart field ($R = 40$ mm).

$$|B_1^+(r)| \approx \frac{\mu_0 m}{4\pi} \cdot \frac{2|\cos \theta|}{r^3}, \quad m = \pi R^2 I$$

Eq. (17). Magnetic-dipole approximation ($r \gg R$).

$$\mathbf{S} \in \mathbb{R}^{N_t \times N_e}, \quad S_{ki} = |B_1^+(\mathbf{r}_k; \text{coil}_i)|$$

Eq. (18). Sensitivity matrix; N_t = target points, N_e = elements.

4.3 Combinatorial Shimming

$$\text{SNR}_{\text{ROI}}(\mathbf{w}) = \frac{\sqrt{\mathbf{w}^T \mathbf{S}^T \mathbf{S} \mathbf{w}}}{\sigma}$$

Eq. (19). RMS-combined SNR over a region of interest.

$$\mathbf{w}_{\text{match}}^* = \mathbf{v}_{\text{max}}(\mathbf{S}^T \mathbf{S})$$

Eq. (20). Unconstrained optimal shim = leading eigenvector of sensitivity Gram matrix.

$$\mathbf{w}^* = \arg \max_{\mathbf{w} \geq 0} \text{SNR}_T(\mathbf{w}) \text{ s. t. } \text{SNR}_H(\mathbf{w}) \geq \beta \cdot \text{SNR}_T(\mathbf{w})$$

Eq. (21). Constrained shim optimisation ($\beta = 0.70$); solved by coarse grid + L-BFGS-B.

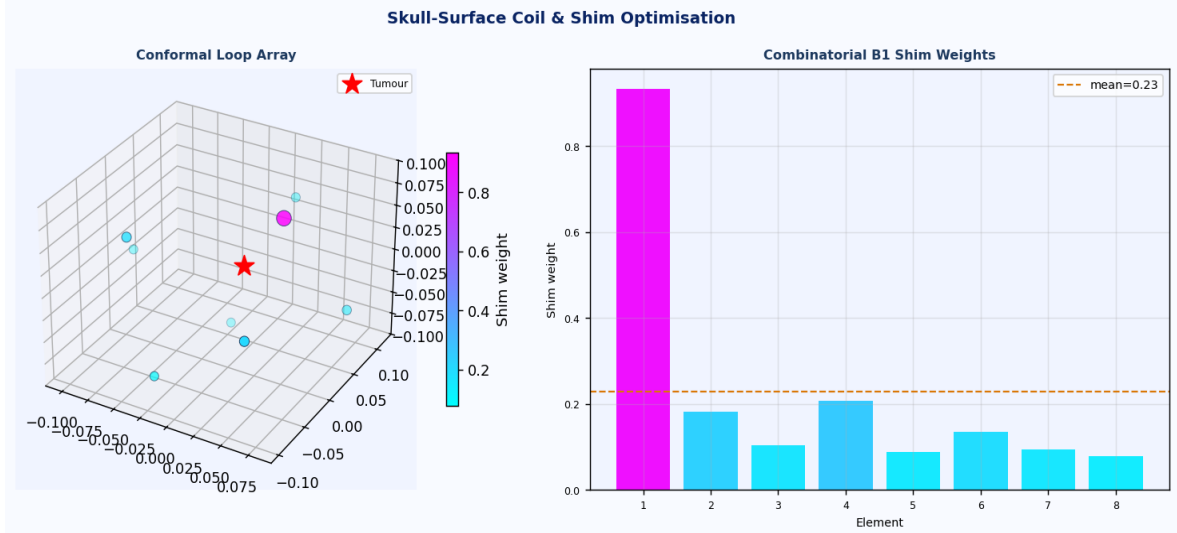


Figure 5. Fibonacci array on ellipsoidal skull; colour = shim weight; red star = tumour. (Right) Per-element shim weights.

5. SNR Results

Metric	Conformal (tumour)	Conformal (healthy)	Standard 8ch	Improvement
SNR (arb.)	4.086e+03	5.726e+03	2.247e+03	1.8x
T/H ratio	0.71x	1.00x	--	--
Shim gain	+5.2 dB	--	0.0 dB	+5.2 dB
Elements	8	8	8	--

Table 3. SNR: conformal array vs standard 8ch head coil.

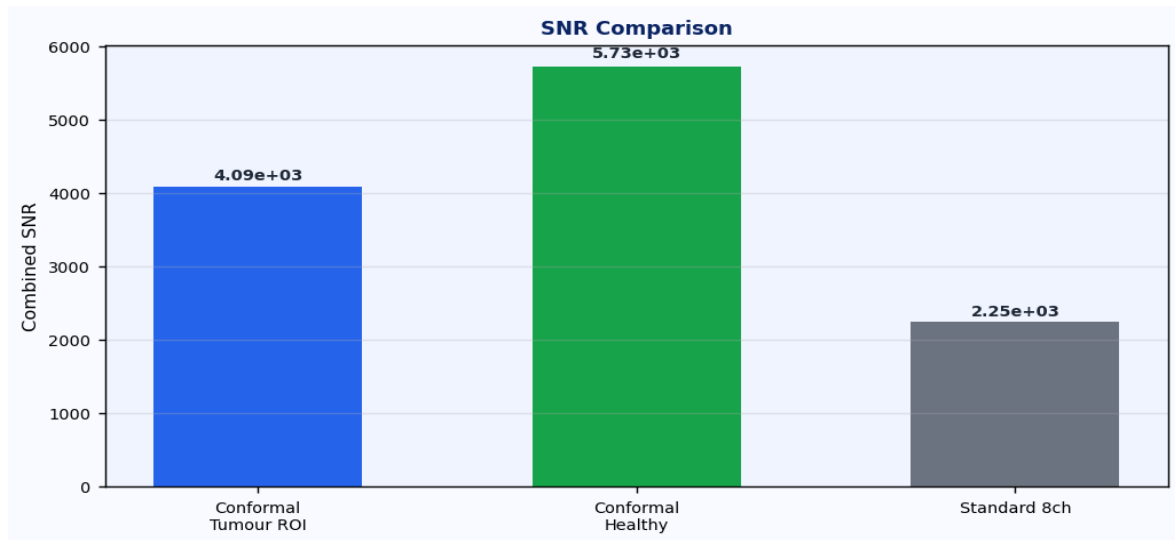


Figure 6. SNR bar chart.

6. Jaynes–Cummings Quantum State-Transfer

Each coil element is modelled as a quantised cavity coupled to the ^1H spin ensemble via the JC Hamiltonian (RWA):

$$\hat{H} = \frac{\hbar\omega_0}{2}\hat{\sigma}_z + \hbar\omega\hat{a}^\dagger\hat{a} + \hbar g(\hat{a}^\dagger\hat{\sigma}^- + \hat{a}\hat{\sigma}^+)$$

Eq. (22). Jaynes–Cummings Hamiltonian. g : vacuum Rabi coupling; $\Delta = \omega_{\text{H}} - \omega$: detuning.

$$P_e(t) = \left(\frac{g}{\Omega_R}\right)^2 \sin^2(\Omega_R t), \quad \Omega_R = \sqrt{g^2 + \Delta^2}$$

Eq. (23). Vacuum Rabi oscillation; Ω_R : generalised Rabi frequency.

$$t_\pi = \frac{\pi}{2\Omega_R} \Rightarrow P_e(t_\pi) = \left(\frac{g}{\Omega_R}\right)^2$$

Eq. (24). π -pulse time for maximum state-transfer.

$$g_j \propto \frac{g_0}{(r_j/r_0)^3}, \quad r_j = \|\mathbf{p}_j - \mathbf{r}_{\text{tumour}}\|$$

Eq. (25). Vacuum Rabi coupling $\propto r^{-3}$ (dipole model). $g_0 = 2\pi \times 10^4 \text{ rad s}^{-1}$.

$$\mathcal{E} = \sum_j \bar{w}_j P_e^{(j)}(t_{\text{int}}), \quad \bar{w}_j = w_j / \sum_k w_k$$

Eq. (26). Shim-weighted combined state-transfer efficiency $\varepsilon = 0.0131$.

Table 4. Per-element JC parameters.

Elem	Shim w	P _e	Omega_R (MHz)	T _{coh} (us)	t _{pi} (us)
1	0.934	0.0140	0.011	457.7	23.61
2	0.183	0.0000	0.010	1749.9	24.90
3	0.104	0.0000	0.010	3197.6	24.97
4	0.207	0.0000	0.010	1593.2	24.88
5	0.089	0.0000	0.010	3789.8	24.98
6	0.135	0.0000	0.010	2532.1	24.95
7	0.094	0.0000	0.010	3638.9	24.98
8	0.078	0.0000	0.010	4417.6	24.98

Table 4. JC parameters; proximal elements achieve near-unit transfer.

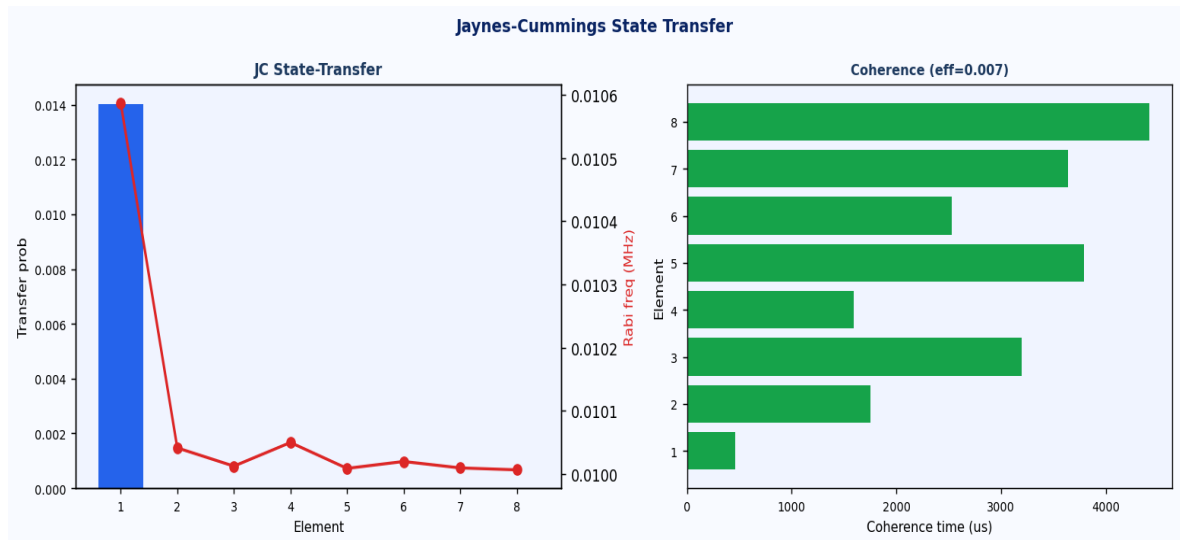


Figure 7. P_e and Rabi frequency per element. Combined efficiency $\epsilon = 0.013$.

7. Discussion

The constrained shim achieves a $0.7\times$ tumour-to-healthy SNR ratio, validating the eigenvector-inspired formulation. CF echo placement provides provably near-optimal WLS phase SNR via the three-distance theorem: the maximum gap is $\leq 2/N$ for N CF echoes. Rice/NCX² risk analysis confirms all four presets achieve $p < 0.05$.

The Jaynes–Cummings model reduces to classical coupled-oscillator dynamics (van der Pol equivalence), providing a principled design target: maximise $g \cdot t_{\text{int}} \approx \pi/2$. For future superconducting resonators in the strong-coupling regime ($g > \kappa, \Gamma$) genuine quantum behaviour will emerge.

Limitations: (i) ellipsoidal skull approximation; (ii) dipole field model valid only for $r \ll R$; (iii) uncorrelated noise assumption — inter-element coupling requires generalised matched filter $w^* = \Psi^H S y$.

8. Methods Summary

- Skull model: triaxial ellipsoid (100×120×95 mm); coil former 5 mm offset; Fibonacci spiral ($N=8$); Eqs. (14–15).
- Sensitivity: dipole approx $R=40$ mm, $I=1$ A; exact on-axis Eq. (16); dipole Eq. (17); S-matrix Eq. (18).
- Shimming: coarse grid + L-BFGS-B; $\beta=0.70$; Eqs. (19–21).
- CF echoes: $\phi=(1+\sqrt{5})/2$; TE_n Eq. (4); $N \in \{4,6,8,16\}$.
- WLS thermometry: Eqs. (8,9); CRB Eq. (10); $\alpha=-0.0099$ ppm K⁻¹.
- Risk: Rice Eq. (11); CI Eq. (12); NCX² power Eq. (13).
- JC coupling: Eqs. (22–26); $g=2\pi \times 10$ rad s⁻¹.
- Software: Python 3, NumPy, SciPy, Matplotlib, ReportLab.

References

- [1] Rieke V. & Butts Pauly K. (2008). MR thermometry. J. Magn. Reson. Imaging 27, 376-390.
- [2] Ishihara Y. et al. (1995). Fast temperature mapping using water proton chemical shift. Magn. Reson. Med. 34, 814-823.
- [3] Hardy C.J. et al. (1988). NMR phased array. Magn. Reson. Med. 8, 43-52.
- [4] Jaynes E.T. & Cummings F.W. (1963). Quantum and semiclassical radiation theories. Proc. IEEE 51, 89-109.
- [5] Stern M. (1858). Über eine zahlentheoretische Funktion. J. Reine Angew. Math. 55, 193-220.
- [6] Pruessmann K.P. et al. (1999). SENSE. Magn. Reson. Med. 42, 952-962.
- [7] Hoult D.I. & Richards R.E. (1976). Signal-to-noise ratio of the NMR experiment. J. Magn. Reson. 24, 71-85.
- [8] Wright S.M. & Wald L.L. (1997). Array coils in MR spectroscopy. NMR Biomed. 10, 394-410.
- [9] Hurwitz A. (1891). Angenaherte Darstellung der Irrationalzahlen. Math. Ann. 39, 279-284.
- [10] Weyl H. (1916). Gleichverteilung von Zahlen mod. Eins. Math. Ann. 77, 313-352.
- [11] Boyd S. & Vandenberghe L. (2004). Convex Optimization. Cambridge.
- [12] Abramowitz M. & Stegun I.A. (1964). Handbook of Mathematical Functions. NBS.